PaperSonar

Grain definitions

Design baseline 1.0 / 25 October 2026

A row must represent the same kind of event or observation throughout its table. The definitions below fix PaperSonar's row meaning, uniqueness and update behavior before extraction or report construction begins.

fact_publication

One row represents one admitted canonical scholarly work.

Primary key: work_key. Only admitted works have publication facts. Two authors, three affiliations or multiple topic assignments do not create extra publication rows. Metadata corrections update the same row after a validated load.

Example: a paper published in 2021, imported twice and observed on three dates still contributes one publication. Its author and topic memberships live in bridges, and its three observation dates belong in the snapshot fact.

fact_work_snapshot

One row represents one admitted work on one successfully published observation day.

Composite primary key: work_key and snapshot_date_key. A successful complete refresh produces one observation for each admitted work. Multiple same-day attempts are reconciled into one validated daily state. A failed load adds none. A later date appends a new row.

Example: citation totals of 18 and 21 on two observation dates imply a recorded increase of 3. Summing 18 and 21 to report 39 citations is invalid. If either count is absent, the change is unavailable. Negative differences are retained as source revisions.

fact_citation

One row represents one distinct directed citing work and cited work pair observed in the retained corpus.

Composite primary key: citing_work_key and cited_work_key. The same reference supplied in repeated loads remains one pair. Different citing works referencing the same target create different pairs. Duplicate references within one source record are deduplicated.

first_seen_date_key is the first local observation. It is not the date the author made the citation. is_current records whether the pair remains present in the latest complete reference list. The target can be a stub, but it must have a dim_work row. Self-links are quarantined.

Related row grains and safeguards

Table group	One row represents	Uniqueness
Topic bridge	One admitted work and one supplied topic	work_key + topic_key
Author bridge	One admitted work and one distinct supplied author	work_key + author_key
Institution bridge	One admitted work and one distinct institution	work_key + institution_key
Work dimension	One canonical work including possible reference stub	work_key with unique openalex_work_id
Topic and entity dimensions	One source entity or controlled category	Surrogate PK and source alternate key
Date dimension	One calendar day or the unknown member	date_key and nullable unique full_date
Run log	One pipeline execution	run_id

Date interpretation

Publication year and observation date answer different questions. Publication trends group by the work's reported publication year. Citation totals describe the source state on the observation day. A source-reported full date may still conceal uncertain precision. Mark it source_reported unless independent evidence supports a day. Use unknown date key 0 when no usable full date exists, retain the known year in dim_work and exclude year-only records from month and elapsed-period analysis.

Population and counting

Reference stubs do not add publication output. Count admitted works before joining multiple bridges. The primary-topic view assigns each work to one topic or unknown. The fractional-topic view assigns 1/N across its N distinct supplied topics. Institutional fractional credit uses the complete deduplicated affiliation set. Those methods answer different questions and must have different report labels.

Change and reproducibility

Current dimensions and bridges use Type 1 updates. Historical citation counts survive in the snapshot fact, but historical classifications are reproduced from archived raw files and corpus versions. Store the classification extraction date in the run manifest. Reports using current classifications must say so. No measure may imply that pre-collection citation snapshots exist.